

Experiment 3: Software Design Principles

Identify stakeholders and perform requirement elicitation for the selected software project using suitable techniques such as interviews, questionnaires, observation, or user personas; prepare user stories and the Requirement Gathering Report.

Aim

To identify stakeholders and perform requirement elicitation for the **AI Interview Preparation Platform** using suitable elicitation techniques, prepare user stories, and develop a Requirement Gathering Report.

Objectives

- Identify key stakeholders of the AI Interview Preparation Platform.
- Apply requirement elicitation techniques.
- Gather functional and non-functional requirements.
- Prepare user stories with acceptance criteria.
- Document the Requirement Gathering Report.

Introduction

Requirement elicitation is the process of discovering stakeholder needs and expectations. The quality of software depends on how accurately user requirements are captured. Common elicitation techniques include interviews, questionnaires, observation, workshops and user personas.

Selected Project

Project Title: AI Interview Preparation Platform

Domain: Artificial Intelligence, Career Development and Education

Stakeholder Identification

Stakeholder	Role	Requirements
Candidate	Uses the platform for interview preparation	Realistic mock interviews, personalized feedback and performance tracking
HR/Recruiter	Uses interview-related requirements and evaluates candidate preparation	Relevant interview questions and skill assessment

Career Counselor	Guides candidates in career and interview preparation	Detailed performance reports and skill-gap information
System Administrator	Manages platform operations	User management, system monitoring and secure operation
AI/ML System	Analyzes interview responses and resumes	Accurate response analysis, feedback and resume evaluation

Requirement Elicitation Techniques

Technique	Purpose	Outcome
Interview	Discuss interview preparation needs with candidates and career professionals	Detailed functional requirements
Questionnaire	Collect feedback from multiple candidates	User expectations and preferences
Observation	Observe how candidates practice interviews	Usability and workflow requirements
User Persona	Model representative candidates	User-centred requirements

Sample Requirement Gathering

Interview Findings:

- Candidates require realistic technical and behavioral mock interviews.
- Candidates need immediate and personalized feedback.
- Interview questions should be relevant to the selected job role.
- The system should evaluate response quality and communication.
- Resume analysis should identify ATS compatibility and skill gaps

Observation Findings:

- Users prefer a simple and easy-to-navigate interface.
- Candidates need clear instructions during mock interviews.
- Frequently used features should be easily accessible.

- Performance results should be presented in an understandable format.

Questionnaire Findings:

- Candidates expect the platform to be available 24×7.
- Personalized feedback is a high priority.
- Users expect accurate resume and interview analysis.
- Candidates prefer progress tracking across multiple practice sessions.

User Personas

Persona	Age	Goal	Pain Point
Final-Year Student	22	Prepare for technical interviews	Lack of real interview practice
Job Seeker	27	Improve interview performance	Limited personalized feedback
Career Switcher	32	Identify skill gaps and role	Uncertainty about required skills

User Stories

ID	User Story	Acceptance Criteria
US1	As a candidate, I want to start an AI mock interview so that I can practice for real interviews.	Feature works correctly and passes functional testing.
US2	As a candidate, I want to practice technical interview questions so that I can improve my technical knowledge.	Feature works correctly and passes functional testing.
US3	As a candidate, I want to practice behavioral questions so that I can improve my communication and confidence.	Feature works correctly and passes functional testing.
US4	As a candidate, I want to receive AI-generated feedback on my answers so that I can identify my weaknesses.	Feature works correctly and passes functional testing.
US5	As a candidate, I want to analyze my resume so that I can check its ATS compatibility.	Feature works correctly and passes functional testing.

Requirement Gathering Report

The elicitation process identified **mock interviews, technical and behavioral interview practice, AI-based feedback, communication analysis, resume analysis, ATS compatibility, skill-gap**

identification and performance tracking as the primary requirements. Stakeholder feedback emphasised ease of use, realistic interview practice, fast feedback, accuracy, reliability and personalized career preparation.

Observation

Using multiple elicitation techniques improves requirement completeness and reduces ambiguity. Interviews, questionnaires, observation and user personas help capture the different needs and expectations of candidates and other stakeholders.

Result

Stakeholders were identified, requirements were elicited using suitable techniques, user stories with acceptance criteria were prepared, and the **Requirement Gathering Report for the AI Interview Preparation Platform** was successfully completed.